POWER PLANT ENGINEERING

Subject Code: 23A03702e

UNIT IV: Hydro Electric Power Plant

1. Hydro Electric Power Plant – Water Power

- ***Hydro power*** converts the ***potential energy of water stored at a height*** (head) into kinetic energy -> mechanical energy in a ***water turbine*** -> electrical energy in a generator.

> *Power P = rho g Q H eta (watts) = 9.81 Q H eta (kW), with Q in m^3/s and H (net head) in metres*

- Example: Q = 20 m^3/s, H = 100 m, overall eta = 0.85 -> P = 9.81 x 20 x 100 x 0.85 = ***16,677 kW = 16.7 MW***
- Energy from hydro is ***renewable, clean, cheap to run*** (no fuel), long life (50–100 years), quick start (2–5 min), and gives ***flood control, irrigation, drinking water, navigation and fisheries***.

Advantages / disadvantages

Advantages	Disadvantages
No fuel cost, low running + maintenance cost	***High capital cost***, long construction time
No pollution; no ash or smoke	Depends on rainfall -> output varies (dry years)
Quick start-up and load pickup (peak + spinning reserve)	Submergence of land, forests, villages (***rehabilitation***)
Long life, high efficiency (85–90%)	Sites are far from the load centre -> long transmission lines
Multi-purpose benefits (irrigation, flood control)	Silting of reservoir; ecological change; earthquakes risk

2. Hydrological Cycle

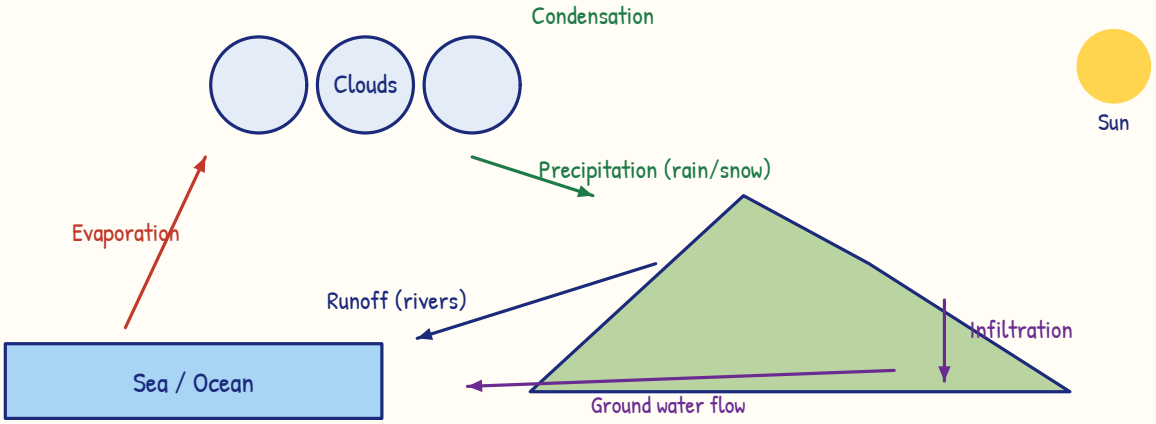


Fig: Hydrological (water) cycle

- Sun's heat -> ***evaporation*** from oceans / lakes / soil, ***transpiration*** from plants -> vapour rises -> ***condensation*** into clouds -> ***precipitation*** (rain, snow, hail) -> ***runoff*** in rivers, ***infiltration*** into ground (ground water) -> back to sea.
- Rainfall in India mainly from the ***south-west monsoon (June – Sept.)*** -- average ~ 1170 mm.

Precipitation measurement

- ***Rain gauge:*** non-recording (Symon's, cylindrical collector) and recording (***tipping bucket, weighing, float type***). The ***average*** rainfall over a basin from many gauges: ***arithmetic mean, Thiessen polygon, isohyetal method*** (most accurate).
- Runoff = Precipitation – Losses (evaporation, infiltration, interception, depression storage). ***Runoff coefficient*** = runoff / rainfall (0.1–0.9; depends on slope, soil, vegetation).

3. Flow Measurement – Streamflow

- ***Rate of flow (discharge) Q = A x V*** (m^3/s). It is needed to find the hydro potential and to design the plant.

Method	Principle
Velocity-area method	Cross-section divided into strips; velocity in each measured by a ***current meter*** (cup / propeller type) at 0.6 depth or 0.2 + 0.8 depth average; Q = sum of (area x velocity)

Method	Principle
Float method	Surface float travels a known distance in time t; mean velocity = 0.85 x surface velocity (approx.)
Weirs and notches	$Q = (2/3)\, C_d\, L\, \sqrt{2g}\, H^{1.5}$ for rectangular weir (small streams); V-notch: $Q = (8/15)\, C_d \tan(\theta/2)\, \sqrt{2g}\, H^{2.5}$
Chemical / salt dilution	Tracer injected, concentration downstream measured; for turbulent mountain streams
Stage-discharge (rating curve)	Gauge height versus measured Q plotted; discharge read from stage thereafter
Ultrasonic / electromagnetic	Modern, continuous measurement

- **Chezy / Manning formula** for uniform flow: $V = (1/n)\, R^{2/3}\, S^{1/2}$ (R = hydraulic radius, S = slope, n = roughness).

4. Drainage Area (Catchment) Characteristics

- **Catchment / drainage basin** = the area from which rainfall drains to a given point of a river. Its **shape, size, slope, soil, vegetation and geology** decide runoff.
- **Size:** larger area -> more runoff, slower response. **Shape:** fan-shaped (round) gives a sharp, high flood peak; long, narrow shape gives a low and flat hydrograph.
- **Slope:** steep -> rapid runoff, less infiltration. **Soil:** impervious (clay, rock) -> high runoff; sandy -> high infiltration.
- **Vegetation / land use:** forests reduce and delay runoff, reduce erosion; urbanisation increases runoff.
- **Drainage density** = total length of streams / area (high density -> fast runoff). **Stream order** (Horton's) and **river slope** are also considered.
- **Evaporation and weather:** rainfall intensity, duration, distribution, temperature, wind.

5. Hydrographs

- A **hydrograph** = plot of **discharge versus time** at a point of a river (stream).

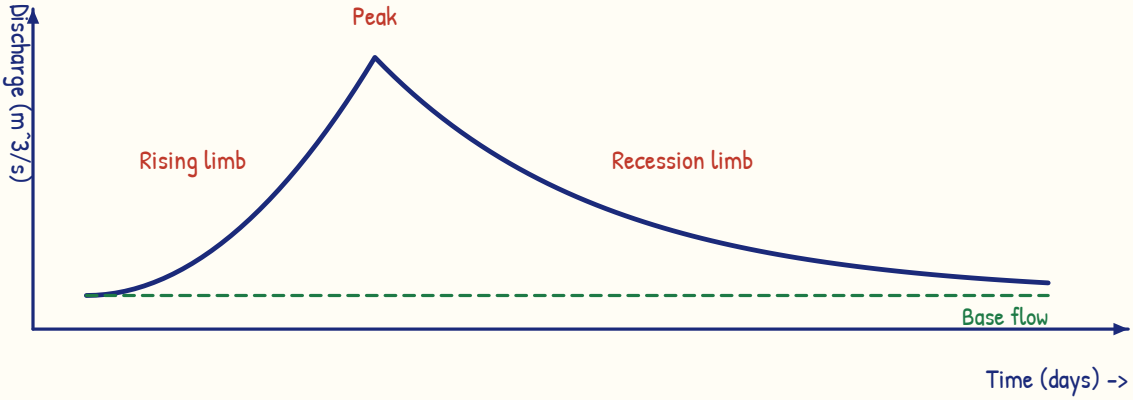


Fig: Flood hydrograph with rising limb, peak and recession limb

- Parts: **rising limb** (surface runoff builds), **crest / peak**, **recession (falling) limb**, and **base flow** (ground water contribution).
- **Unit hydrograph:** direct runoff hydrograph from **1 cm of effective rainfall** of a given duration (D hours) spread uniformly over the basin. Used (linear superposition) to predict floods from a storm.
- **Hyetograph:** rainfall intensity versus time (bar graph).

Flow duration curve

- Plot of discharge (Q) against the **percentage of time** that this discharge is equalled or exceeded. The area under the curve gives the volume of water / **power potential**. Used for choosing plant capacity, **firm (dependable) power** (e.g. Q_{90} or Q_{75}), and the type of plant (run-of-river or storage).

Mass curve (Ripl's method)

- Plot of **cumulative inflow volume versus time**. The slope = flow rate. A line of constant demand drawn tangent to the crests gives the **required reservoir capacity** (maximum vertical distance between the demand line and the mass curve).

Example 1: Reservoir capacity

- Mass curve for the lean months: inflow 40, 30, 20, 25 Mm^3; steady demand 35 Mm^3 each month.
- Deficit: 0, 5, 15, 10 -> cumulative deficit 0, 5, 20, 30 Mm^3. Reservoir capacity required = **30 Mm^3**.

6. Storage and Pondage

- **Storage:** large reservoir that holds **excess flow of the rainy season** to be used in the dry season (seasonal regulation). Plants with storage are mostly **peak or base** load plants and give a steady output.
- **Pondage:** small storage behind a **run-of-river** plant weir / barrage which stores water during **off-peak hours of the day (night)** to meet the **peak demand** (daily / weekly regulation).
- **Plant use factor** = actual energy generated / (installed capacity x hours of operation). Pondage allows a run-of-river plant to run at higher capacity during peak hours.
- **Reservoir zones:** dead storage (silt), live (useful) storage, flood control storage (above full reservoir level).

Level	Meaning
Minimum Drawdown Level (MDDL)	Lowest level for power; below this dead storage
Full Reservoir Level (FRL) / Normal Pool	Maximum level for normal storage
Maximum Water Level (MWL)	Highest level during design floods (with spillway)
Free board	Height of dam top above MWL (waves, safety)

- **Reservoir sedimentation:** silt reduces capacity (0.5–1% per year); controlled by catchment treatment, silt sluices, dredging, trap efficiency, bypass.

7. Classification of Dams

- **Dam** = a barrier built across a river to create a reservoir and **head**. Choose by site, foundation, material, climate, cost.

Type	Features
Gravity dam	Masonry / concrete; stability by its **own weight**; needs good rock foundation; most common for 50+ m; Bhakra, Nagarjuna Sagar (masonry)

Type	Features
Arch dam	Curved upstream in plan; transmits water load to abutments by arch action; narrow gorge, strong rock sides; less concrete; Idukki
Buttress dam	Slanting deck supported by buttresses; saves concrete; wide valley; skilled labour
Embankment dam (earth / rock-fill)	Earth or rock; good for weak foundations; cheaper; needs a **spillway** separate; Tehri (earth + rock-fill)
Composite dam	Concrete + embankment parts

Forces on a gravity dam

- Water pressure (horizontal), **self weight**, **uplift pressure**, silt pressure, wave, ice, earthquake, wind.
- Checks: no **overturning**, **sliding** (factor of safety > 1.5 with shear friction), no **tension** in concrete, compressive stress within limit; resultant must lie in the **middle third** of the base.

8. Spillways

- A **spillway** safely discharges surplus / **flood** water from the reservoir to the downstream river **without damaging the dam**.

Spillway	Description
Overflow (ogee)	Crest shaped like the lower nappe of a jet over a sharp weir; on **concrete dams**; with gates
Chute (trough / open channel)	Steep open channel to the side of the earth dam
Side-channel	Flow goes over crest, then turns 90 deg in a side channel parallel to the crest
Shaft (morning glory)	Funnel-shaped inlet, vertical shaft, then tunnel; for narrow gorges
Siphon	Automatic; primes and discharges at the full level; for small variation

Spillway	Description
Tunnel / Conduit	Water carried through hill or dam body

- **Gates:** vertical lift, **radial (Tainter) gates** (most used), drum, rolling. **Energy dissipators** (stilling basin, flip bucket, hydraulic jump) at the toe to reduce scour.
- Design flood: standard project flood (SPF) / **probable maximum flood (PMF)** based on rainfall.

9. Hydro Power Plant – Types and Classification

- **Based on water use / availability:**
  - **Run-of-river:** no or little storage; output varies with flow; low head; base plants on rivers; small / mini / micro hydro.
  - **Storage (reservoir):** large dam; steady operation; for base, peak loads.
  - **Pumped storage:** water pumped up at off-peak and reused at peak.
  - **Tidal:** see Unit V.
- **Based on head: Low (< 15 m), medium (15–70 m), high (> 70 m)** head plants.
- **Based on load:** base load, peak load plants.
- **Based on capacity:** pico (< 5 kW), micro (< 100 kW), mini (< 2 MW), small (< 25 MW), large (> 25 MW).
- **Based on location of power house: surface, semi-underground** and **underground** power house; **dam-toe, canal-side** and **conduit** type.

Turbine selection (summary)

Turbine	Type	Head	Specific speed
Pelton wheel	Impulse	High (> 250 m), small Q	10–35
Francis	Reaction (radial / mixed)	Medium (40–400 m)	60–300
Kaplan / propeller	Reaction (axial)	Low (< 40 m), large Q	300–1000

- Pelton: jets strike **buckets**; nozzle with spear controls flow. Francis: spiral casing, guide vanes, draft tube. **Kaplan:** adjustable runner blades -> good part-load efficiency.

Water turbines in more detail

- **Pelton wheel (impulse):** all head is converted to kinetic energy in the nozzle; jet velocity $V_1 = C_v \sqrt{2gH}$ (C_v = 0.97–0.99); the wheel works at **atmospheric pressure**; bucket speed $u = 0.45 \times V_1$ (ideal **0.5 V_1**); runs 12–35 buckets; needs **no draft tube**; efficiency 85–90%. Governing by **spear (needle) valve** and **deflector** (to avoid water hammer).
- **Francis (reaction):** water enters through **spiral casing** -> **stay vanes** -> adjustable **guide vanes** -> runner (radial inward, mixed-flow exit) -> **draft tube**. Part of the head is pressure head. Efficiency up to 93–95%. Governing by guide vane opening.
- **Kaplan (axial reaction):** runner like a ship propeller with **adjustable blades** (and adjustable guide vanes: **double regulation**) -> high efficiency even at part load; for large discharge, low head (bulb turbines for 2–15 m heads).

> Specific speed $N_s = N \sqrt{P} / H^{5/4}$ (N in rpm, P in kW, H in m)

- **Unit quantities:** $N_u = N / \sqrt{H}$; $Q_u = Q / \sqrt{H}$; $P_u = P / H^{3/2}$ -- used to predict the performance at a different head (**similarity / model laws**).
- **Cavitation:** formation and collapse of vapour bubbles where local pressure falls below vapour pressure -> pitting, noise, vibration, fall in efficiency. Controlled by limiting suction (draft) head, proper runner design, use of **stainless steel**, and the **Thoma cavitation factor** $\sigma = (H_{atm} - H_{vap} - H_s) / H$.
- **Draft tube:** diverging tube (conical / elbow) which **recovers kinetic energy** and lets the runner be above tail-water level; efficiency 70–90%.

Example 3: Pelton wheel power

- A jet of water 0.1 m^3/s under a net head of 400 m strikes a Pelton wheel (C_v = 0.98, overall efficiency 0.85).
- Jet velocity = $0.98 \sqrt{2 \times 9.81 \times 400} = 0.98 \times 88.6$ = **86.8 m/s**
- Shaft power = $9.81 \times 0.1 \times 400 \times 0.85$ = **333.5 kW**

Governing of hydro turbines

- Mechanical-hydraulic or electro-hydraulic **governor** with speed sensor (flyball / tachometer), servo-motor and oil pressure unit. Speed regulation 3–5%.
- **Water hammer** = pressure rise when flow is stopped suddenly (Joukowsky): delta p = rho c V; closing time of guide vanes is chosen long enough (> 2L/c) and a **surge tank** or **pressure relief valve** is provided.

Power estimation with flow duration

Example 4: Energy from a run-of-river site

- Available head 30 m, plant efficiency 0.85. A flow duration curve gives: 40 m^3/s for 30% of year, 25 m^3/s for 40%, 10 m^3/s for 30%.
- Power: 10,006 kW, 6254 kW and 2502 kW respectively; Energy per year = 8760 x (0.3 x 10,006 + 0.4 x 6254 + 0.3 x 2502) = **8760 x 6254 = 54.8 x 10^6 kWh**
- Average load = 6254 kW; with plant of 10 MW -> **capacity factor 62.5%**; firm power ~ 2.5 MW.

10. Typical Layouts and Plant Auxiliaries

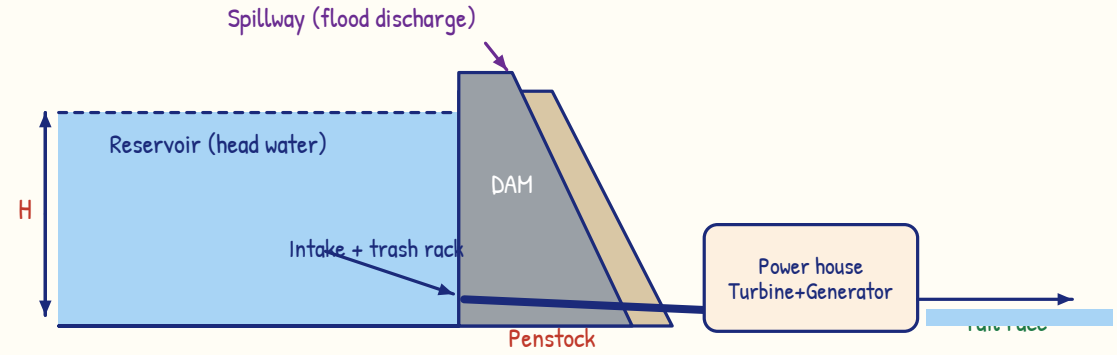


Fig: Layout of a storage-type hydro plant (dam, penstock, power house)

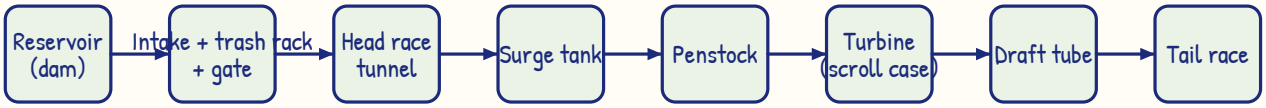


Fig: Components in the water path

- **Dam and reservoir** hold water and give head. **Intake / trash rack** stops debris; gates control flow.

- **Penstock:** steel / RCC pipe carrying water at high pressure to the turbine; **water hammer** is controlled by the surge tank and relief valves.
- **Surge tank:** an open chamber connected to the penstock which **absorbs the pressure surge** from sudden load change (reduces water hammer) and provides water during rapid load rise. Types: simple, restricted orifice, differential (Johnson), closed.
- **Power house:** houses turbines, generators, control room, **EOT crane**, switchyard. **Draft tube** recovers the exit kinetic energy and allows the turbine to be above tail water. **Tail race** returns the water to the river.

Plant auxiliaries

- **Governor** (oil pressure actuated; changes gate / needle opening to hold speed), **inlet valve (butterfly / spherical)**, **cooling water system** for generator and bearings, **lubrication** and **brake system**.
- **Excitation system**, transformers, switchyard (circuit breakers, isolators, lightning arresters), **DC battery** supply, fire protection, drainage and dewatering pumps, ventilation (for underground plants), cranes.
- **Trash-rack cleaning machine, fish ladder / pass**, cathodic protection, instrumentation and SCADA.

II. Plant Operation

- **Start-up:** open inlet valve bypass -> fill the penstock and scroll case -> open guide vanes -> accelerate to rated speed -> excite generator -> **synchronise** with grid -> load by opening the guide vanes.
- **Shut-down:** unload, trip the breaker, close guide vanes, close inlet valve (and the intake gate for maintenance); brakes applied at low speed (< 30%).
- **Control:** The **governor** keeps speed (frequency) constant. **Load-frequency control (AGC)** and **reactive power / voltage control** by the AVR.
- **Operation in the grid:** hydro units pick up load in a few minutes -> used as **peaking stations** and **spinning reserve**, **frequency regulation**; thermal / nuclear take the base load.
- **Operating rules:** avoid rough zones (vibration, cavitation), monitor bearing temperatures, vibration, air gap, oil levels; **cavitation** damage at runner blades is prevented by correct setting of turbine and by using stainless steel.

- Maintenance: annual overhaul in dry period; silt erosion repair; inspection of dam (***dam safety***) and instruments (piezometers, seepage, tilt meters).

12. Pumped Storage Plants

- Used with thermal / nuclear base-load plants. ***Off-peak*** surplus power (night) is used to ***pump water from a lower reservoir to an upper reservoir***; during ***peak*** hours the water flows down through turbines to generate power.

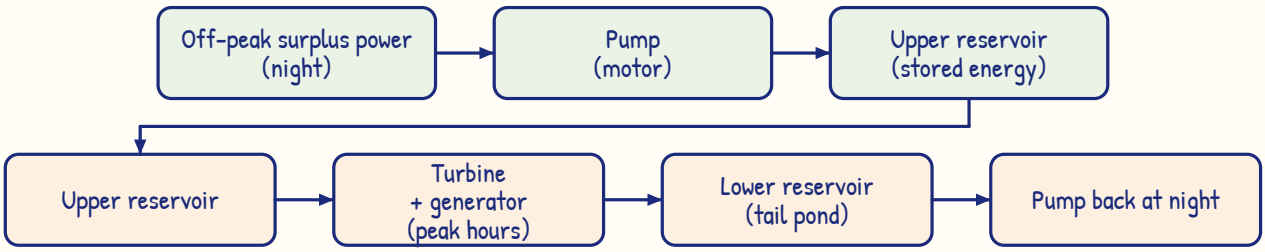


Fig: Working of pumped storage plant

- Arrangements: ***two-reservoir*** type (separate pump and turbine) or ***reversible pump-turbine*** with motor-generator (***most economical***; Francis type reversible).
- ***Overall (cycle) efficiency 70–80%*** (pump 90% x turbine 90% x losses). It gives ***load levelling***, improves system load factor, and acts as a ***huge battery***; useful with ***solar / wind*** integration.
- India: Kadamparai, Bhira, Srisailam LBPH, ***Purulia (900 MW)***, Tehri PSP, Kadana. Potential about 96 GW.

Example 2: Pumped storage

- Upper reservoir 100 m above lower; volume 5 x 10^6 m^3. Energy stored = rho g V H = 1000 x 9.81 x 5 x 10^6 x 100 = 4.9 x 10^{12} J = ***1362 MWh***.
- With round-trip efficiency 75%, recoverable energy = ***~ 1020 MWh***

13. Hydro Power in India and Small Hydro

- ***Major stations:*** Bhakra-Nangal (1325 MW, Gobind Sagar), Tehri (1000 MW + 1000 MW PSP), ***Sardar Sarovar***, Koyna (Maharashtra), Nagarjuna Sagar, Srisailam, Idukki (arch dam), Sharavathi, Subansiri, Kishanganga.
- Share of hydro in installed capacity ~ ***11%*** (earlier ~ 40–45%); it is important as ***peaking*** and for ancillary services. Development agencies: NHPC, SJVNL,

state boards, THDC.

- **Small hydro (SHP):** up to **25 MW**; mini (100 kW – 2 MW), micro (up to 100 kW); simple run-of-river, **no large dam**, little displacement; suitable for hilly, remote areas; classes by head: ultra low (< 3 m), low (3–30), medium, high.
- Pico / micro hydro: **water mills (gharats)** modernised, crossflow (**Banki**) and Turgo turbines, induction generators; cost Rs 1–1.5 lakh per kW.

Hydro-thermal co-ordination

- Hydro plants take **peak load** and thermal / nuclear plants run at constant base load -> system operates economically.
- In a **wet season** run hydro as base to use the water that otherwise spills; in a **dry season** keep hydro for peak hours only.
- Hydro units are **excellent for frequency control** because of fast loading rate (**~ 20–50% of capacity per minute**).

14. Short Questions (exam revision)

Q & A

- **Why is a surge tank provided?** To relieve water hammer pressure in the penstock and to supply extra water when the load increases suddenly.
- **Difference between storage and pondage?** Storage = seasonal, large reservoir; pondage = small, daily / weekly regulation for run-of-river plants.
- **What is the unit hydrograph?** Hydrograph of direct runoff resulting from unit (1 cm) depth of effective rainfall uniformly spread on the basin in specified duration.
- **Why Kaplan for low head?** Large flow requires large flow area; axial flow runner with adjustable blades gives high efficiency for large Q at low head.
- **What is a draft tube?** Gradually expanding tube connecting the runner exit to tail race; converts kinetic energy to pressure, raising the effective head.
- **What is the firm power?** Power continuously available from a plant at all times (~ 90–95% of the time).

- ***Run-of-river advantage?*** Low cost, no submergence; drawback: output varies with river flow.

Quick revision

- P (kW) = 9.81 Q H eta. Pelton: high head; Francis: medium; Kaplan: low head.
- Storage = seasonal; pondage = daily. Surge tank = relieves water hammer. Spillway = flood escape.
- Pumped storage: pump at night, generate at peak; efficiency 70–80%.